

Playwright Capstone Project Planner

Project Name: Enterprise E-Commerce Automation Testing Framework

Selected Website: <https://automationexercise.com/>

This document outlines the architecture, implementation strategy, execution roadmap, planned modules, and automation goals for the Playwright Capstone Project using enterprise-grade automation practices.

1. Project Objectives

- Minimum 120 automated test cases
- Minimum 8 services/modules
- Playwright automation framework
- Allure and HTML reporting
- Reusable Page Object Model (POM) structure
- API testing and mocking support

2. Services & Test Case Breakdown (8 × 15 = 120 Test Cases)

#	Service	Key Test Scenarios	Cases
1	Authentication	Valid login, invalid login, logout, registration, password reset, session persistence, token expiry, locked account, guest access, remember me, brute force, empty fields, SQL injection attempt, redirect after login.	15
2	Product Catalog	Search by name, filter by category, sort by price, product detail page, image display, out-of-stock label, pagination, breadcrumb navigation, related products, URL routing, add to favourites, category count, page load.	15
3	Cart	Add item, remove item, update quantity, empty cart state, price total, apply coupon, cart persistence, max quantity limit, cart badge update, duplicate item handling, continue shopping, tax display, checkout redirect.	15
4	User Profile	Update name, update email, change password, order history, validation errors, phone number update, success confirmation, cancel changes, password strength validation, email format validation, session after update.	15

#	Service	Key Test Scenarios	Cases
5	Address & Shipping	Add address, edit address, delete address, invalid zip code, shipping selection, multiple addresses, default address, delivery estimation, required fields, invalid phone number, country dropdown validation.	15
6	Payment	Valid payment, invalid card, expired card, invalid CVV, payment confirmation, failed payment handling, multiple payment attempts, transaction validation, payment redirect, secure payment icon validation.	15
7	Customer Support	Contact form submission, invalid email, attachment upload, success message, reset form, FAQ navigation, long message handling, mandatory fields, ticket generation, support response confirmation.	15
8	API Testing	GET products API, POST login API, invalid request, status code validation, response schema validation, authentication token validation, unauthorized access, invalid endpoint, API consistency validation.	15

3. Planned Features

- Page Object Model (POM)
- Data-driven testing
- Parallel execution
- Annotations (skip, only, fixme)
- Screenshot on failure
- Trace and video capture
- API mocking using page.route()
- Environment variable handling
- HTML and Allure reports
- Reusable utility functions

4. Execution Timeline

Day	Planned Work
Day 1	Framework setup and repository initialization
Day 2	Authentication and Product modules

Day 3	Cart and Checkout modules
Day 4	Profile and Shipping modules
Day 5	Customer support and API testing
Day 6	Reporting, screenshots, traces, debugging
Day 7	Documentation and final submission

5. Expected Outcome

The final project will demonstrate enterprise-grade Playwright automation concepts including reusable framework architecture, robust locators, reporting, API testing, screenshots, synchronization handling, and maintainable automation workflows.